

Volatility in Real Time

Leverage, feedback, or both?

FEB22009-25 Financial Case Study

Rutger-Jan Lange

Evgenii Vladimirov

Luuk de Wit

The basic question

In equity markets, large negative returns often come together with large increases in volatility.

Story	Simple idea	Timing
Leverage	bad returns raise volatility	return first, volatility later
Feedback	higher volatility pushes prices down	volatility and return same day
Both	both channels matter	lagged and same-day effects

Main case question: can real-time GARCH models explain and forecast volatility better than classic GARCH models?

Data students use

- ▶ Baseline file: `data.xlsx`.
- ▶ Asset: daily E-mini S&P 500 futures.
- ▶ Sample: 30 September 2009–31 March 2026.
- ▶ Return:

$$r_t = 100 \log \left(\frac{\text{Close}_t}{\text{Close}_{t-1}} \right).$$

- ▶ Realised variance RV5_SS is used as the ex-post volatility proxy.
- ▶ VIX is used as a market-based volatility benchmark.

First task: plot the data and describe the sample before estimating models.

Models in the case

	Symmetric	Asymmetric
Classic update	GARCH	GJR-GARCH
Real-time update	RT-GARCH	RT-GJR-GARCH

- ▶ **Classic GARCH:** yesterday's information determines today's volatility.
- ▶ **Real-time GARCH:** today's return can also help filter today's volatility.
- ▶ **Asymmetry:** negative and positive shocks may have different volatility effects.

The model comparison is a way to compare the economic stories: leverage, feedback, or both.

What students need to produce

1. Describe the data: plots, units, scaling, and summary statistics.
2. Estimate the four GARCH-type models.
3. Compare in-sample fit using likelihoods and information criteria.
4. Create one-day-ahead volatility forecasts.
5. Compare forecasts with realised variance and the VIX benchmark.
6. Explain what the results imply for leverage, feedback, or both.

The final report should not just rank models. It should use the **statistical analysis** to tell a clear **story**: leverage, feedback, or both?

Deliverables

Deliverables

- ▶ Proposal: up to two pages.
- ▶ Final report: up to 5,000 words or 12 pages.
- ▶ Report should be written in $\text{\LaTeX}$.

Important:

- ▶ tell a story;
- ▶ avoid look-ahead bias;
- ▶ explain modelling choices;
- ▶ interpret parameters economically;
- ▶ support your story with statistical analyses.

Goal: Give an evidence-based answer to the case question.